

Mathematical Proof II: Exploring a Problem¹

1 THE GOAL

In the previous session, we practiced how to **deconstruct** a mathematical statement:

What is given? What must be shown? What definitions matter?

Today we focus on the next question:

Once I understand the problem, how do I find a path to the proof?

This is often the hardest stage of mathematical problem solving.

A finished proof may look like

$$A \implies B \implies C \implies D.$$

But when solving the problem, we usually begin with something more like

$$A \implies ? \qquad ? \implies D.$$

The objective today is to develop systematic ways to explore the gap.

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2 DISCOVERY IS NOT THE SAME AS PRESENTATION

A proof is usually written in a clean logical order:

$$A \implies B \implies C \implies D.$$

But that does not mean the proof was discovered in that order.

Often, we begin with the conclusion D and ask:

What would be enough to make D true?

This may lead us backward:

$$D \Leftarrow C \Leftarrow B.$$

At the same time, we can work forward from the assumptions:

$$A \implies B.$$

Eventually the two directions meet.

$$A \longrightarrow B \longrightarrow C \longrightarrow D$$

Assumptions

Goal

You do not need to discover a proof in the same order
in which you eventually write it.

3 STRATEGY 1: WORK FORWARD FROM THE ASSUMPTIONS

A natural way to begin exploring is to ask:

What can I immediately conclude from what I am given?

This means unpacking assumptions and following their consequences.

Consider:

Proposition 1

Suppose A is a positive definite matrix. Show that A is nonsingular.

We first deconstruct the goal.

To show that A is nonsingular, it is enough to show

$$Ax = 0 \implies x = 0.$$

Now work forward from the assumption $Ax = 0$.

If

$$Ax = 0,$$

then multiplying by $x^\top$ gives

$$x^\top Ax = 0.$$

But positive definiteness tells us that if $x \neq 0$, then

$$x^\top Ax > 0.$$

These two statements cannot both hold.

So the forward chain is

$$Ax = 0 \implies x^\top Ax = 0 \implies x = 0.$$

Solution

Suppose $Ax = 0$. Then

$$x^\top Ax = x^\top 0 = 0.$$

If $x \neq 0$, positive definiteness would imply

$$x^\top Ax > 0,$$

a contradiction.

Therefore $x = 0$. Hence the null space of A contains only the zero vector, so A is nonsingular.

4 EXERCISE: GENERATE CONSEQUENCES

For each assumption below, write down at least two consequences that might be useful in a proof.

Do not try to solve a particular theorem yet. The objective is simply to practice extracting usable information.

(a) f is strictly increasing.

(b) f is strictly concave.

(c) $C \subseteq \mathbb{R}^n$ is convex.

(d) $x^* \in \operatorname{argmax}_{x \in C} f(x)$.

(e) A is positive definite.

5 STRATEGY 2: WORK BACKWARD FROM THE GOAL

Sometimes the assumptions do not immediately suggest a path.

Then ask:

What would be sufficient to prove the conclusion?

Consider the inequality

Proposition 1

For all $x, y \in \mathbb{R}$,

$$2xy \leq x^2 + y^2.$$

Suppose we do not immediately know how to prove this.

Start from the desired conclusion:

$$2xy \leq x^2 + y^2.$$

Move everything to one side:

$$0 \leq x^2 - 2xy + y^2.$$

Recognize the expression:

$$0 \leq (x - y)^2.$$

Now we have reached something obviously true.

The proof was *discovered backward*:

$$2xy \leq x^2 + y^2 \iff 0 \leq x^2 - 2xy + y^2 \iff 0 \leq (x - y)^2.$$

But we present it forward.

Solution

Since

$$(x - y)^2 \geq 0,$$

we have

$$x^2 - 2xy + y^2 \geq 0.$$

Rearranging,

$$2xy \leq x^2 + y^2.$$

Backward reasoning is often useful for discovering a proof.
The final proof should still be written in a logically valid forward order.

6 EXERCISE: WORK BACKWARD

Try to work backward from each desired conclusion until you reach a statement that is obviously true or follows directly from the assumptions.

6.1 A Basic Inequality

Show that for $x > 0$,

$$x + \frac{1}{x} \geq 2.$$

Start from the goal:

$$x + \frac{1}{x} \geq 2.$$

What equivalent statement would be easier to recognize?

6.2 Distance from the Mean

Show that for any $x, y \in \mathbb{R}$,

$$\left(\frac{x + y}{2} \right)^2 \leq \frac{x^2 + y^2}{2}.$$

Work backward from the desired inequality.

7 STRATEGY 3: WORK FORWARD AND BACKWARD AT THE SAME TIME

Often the most useful approach is to explore from both ends.

Suppose we want to prove:

Proposition 1

Let $f : C \rightarrow \mathbb{R}$ be concave, where $C \subseteq \mathbb{R}^n$ is convex. If $x^*, y^* \in \operatorname{argmax}_{x \in C} f(x)$, then every convex combination

$$z_\lambda = \lambda x^* + (1 - \lambda)y^*$$

with $\lambda \in [0, 1]$ is also a maximizer.

Work forward from the assumptions

Since C is convex,

$$z_\lambda \in C.$$

Since f is concave,

$$f(z_\lambda) \geq \lambda f(x^*) + (1 - \lambda)f(y^*).$$

Since x^* and y^* are maximizers,

$$f(x^*) = f(y^*) = M.$$

Therefore

$$f(z_\lambda) \geq M.$$

Work backward from the goal

To show that z_λ is a maximizer, we need to show

$$f(z_\lambda) = M.$$

Because M is already the maximum possible value, feasibility of z_λ implies

$$f(z_\lambda) \leq M.$$

So we have

$$f(z_\lambda) \geq M$$

from concavity, and

$$f(z_\lambda) \leq M$$

from maximality.

Hence

$$f(z_\lambda) = M.$$

Solution

Let

$$z_\lambda = \lambda x^* + (1 - \lambda)y^*$$

for some $\lambda \in [0, 1]$.

Since C is convex, $z_\lambda \in C$.

Because f is concave,

$$\begin{aligned} f(z_\lambda) &\geq \lambda f(x^*) + (1 - \lambda)f(y^*) \\ &= \lambda M + (1 - \lambda)M \\ &= M. \end{aligned}$$

But M is the maximum value of f on C , so because $z_\lambda \in C$,

$$f(z_\lambda) \leq M.$$

Therefore

$$f(z_\lambda) = M,$$

so z_λ is also a maximizer.

8 LOOK FOR AN INTERMEDIATE TARGET

A difficult conclusion is often easier to reach through an intermediate statement.

Instead of trying to prove

$$A \implies D$$

directly, search for something C such that

$$A \implies C \quad \text{and} \quad C \implies D.$$

Consider:

Proposition 1

Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ is strictly increasing. Show that if $f(x) \leq f(y)$, then

$$x \leq y.$$

The goal is

$$x \leq y.$$

One way to explore backward is to ask:

What would contradict $x \leq y$?

That would be

$$x > y.$$

Now the assumption of strict monotonicity becomes useful:

$$x > y \implies f(x) > f(y),$$

which contradicts

$$f(x) \leq f(y).$$

The useful intermediate statement was not obvious from the original problem. It emerged by asking what failure of the conclusion would imply.

9 STRATEGY 4: TRY SIMPLE EXAMPLES

When you are unsure whether a statement is true, or unsure what might drive the proof, try simple examples.

Examples can help you:

- understand what the statement is claiming;
- identify which assumptions matter;
- guess what the proof should use;
- discover that the statement is false.

Consider:

If f and g are increasing functions, is

$$h(x) = f(x)g(x)$$

necessarily increasing?

Before trying to prove it, test examples.

Take

$$f(x) = x, \quad g(x) = x.$$

Then

$$h(x) = x^2,$$

which is not increasing on all of $\mathbb{R}$.

So the original statement is false.

This saves us from searching for a proof that cannot exist.

10 EXERCISE: TEST THE CLAIM

For each statement below:

1. test one or two simple examples;
2. decide whether the statement seems true or false;
3. if false, find a counterexample;
4. if true, identify what feature of your examples suggests a proof.

10.1 Products of Increasing Functions

If f and g are strictly increasing, then fg is strictly increasing.

10.2 Sums of Convex Functions

If f and g are convex, then $f + g$ is convex.

10.3 Maximum of Convex Functions

If f and g are convex, then

$$h(x) = \max\{f(x), g(x)\}$$

is convex.

11 STRATEGY 5: SEARCH FOR A COUNTEREXAMPLE

A proof problem does not always come with the guarantee that the statement is true.

Even when the statement is true, trying to break it can reveal why the assumptions are necessary.

Consider:

Suppose $f : C \rightarrow \mathbb{R}$ is concave. Then f has a unique maximizer.

This sounds plausible, but it is false.

Take

$$f(x) = 0 \quad \forall x \in [0, 1].$$

The function is concave, but every point in $[0, 1]$ is a maximizer.

This tells us something useful:

concavity is not enough for uniqueness.

Strict concavity is the stronger condition that rules out multiple distinct maximizers.

12 COUNTEREXAMPLES AS DIAGNOSTIC TOOLS

A good counterexample does more than show that a claim is false.

It can reveal which assumption is missing.

Suppose someone claims:

If $f'(x^*) = 0$, then x^* is a maximizer of f .

Try

$$f(x) = x^3.$$

Then

$$f'(0) = 0,$$

but 0 is not a maximizer.

Or try

$$f(x) = x^2.$$

Again,

$$f'(0) = 0,$$

but 0 is a minimizer.

So the first-order condition by itself is not sufficient.

What additional structure would help?

For example, if f is concave, then a stationary point is a global maximizer under appropriate differentiability conditions.

Thus counterexample search can lead to a better theorem.

13 EXERCISE: REPAIR THE STATEMENT

For each false statement:

1. find a counterexample;

2. explain exactly why the original argument fails;
3. suggest an additional assumption that would make the statement true.

13.1 Stationary Point

If

$$f'(x^*) = 0,$$

then x^* maximizes f .

13.2 Concavity and Uniqueness

If f is concave, then f has at most one maximizer.

13.3 Positive Semidefinite Matrices

If A is positive semidefinite and

$$x^\top Ax = 0,$$

then $x = 0$.

14 STRATEGY 6: REWRITE THE PROBLEM

Sometimes the most useful exploration step is to rewrite the statement in a different but equivalent form.

Consider:

Proposition 1

For $a, b > 0$,

$$\frac{a+b}{2} \geq \sqrt{ab}.$$

If the square root makes the expression difficult to work with, note that both sides are nonnegative, so we can square the inequality:

$$\left(\frac{a+b}{2}\right)^2 \geq ab.$$

Multiplying by 4,

$$(a+b)^2 \geq 4ab.$$

Expanding,

$$a^2 - 2ab + b^2 \geq 0,$$

or

$$(a-b)^2 \geq 0.$$

Again, the original conclusion becomes easy after rewriting it into a more familiar form.

15 A FIRST-YEAR MICRO EXAMPLE: QUASICONCAVITY

Consider a function $u : C \rightarrow \mathbb{R}$, where $C \subseteq \mathbb{R}^n$ is convex.

One definition of quasiconcavity is:

Definition 1 (Quasiconcavity)

A function $u : C \rightarrow \mathbb{R}$ is quasiconcave if for every $x, y \in C$ and $\lambda \in [0, 1]$,

$$u(\lambda x + (1 - \lambda)y) \geq \min\{u(x), u(y)\}.$$

Suppose we are asked:

Proposition 1

Every concave function is quasiconcave.

Rather than immediately writing a proof, explore the gap.

What does concavity give us?

$$u(\lambda x + (1 - \lambda)y) \geq \lambda u(x) + (1 - \lambda)u(y).$$

What do we want?

$$u(\lambda x + (1 - \lambda)y) \geq \min\{u(x), u(y)\}.$$

The missing link is therefore:

$$\lambda u(x) + (1 - \lambda)u(y) \geq \min\{u(x), u(y)\}.$$

Is that true?

Yes. A weighted average of two numbers lies between them.

So we have found the intermediate statement.

$$\text{concavity} \implies u(z) \geq \lambda u(x) + (1 - \lambda)u(y) \implies u(z) \geq \min\{u(x), u(y)\}.$$

Solution

Take arbitrary $x, y \in C$ and $\lambda \in [0, 1]$.

By concavity,

$$u(\lambda x + (1 - \lambda)y) \geq \lambda u(x) + (1 - \lambda)u(y).$$

Because

$$\lambda u(x) + (1 - \lambda)u(y) \geq \min\{u(x), u(y)\},$$

we obtain

$$u(\lambda x + (1 - \lambda)y) \geq \min\{u(x), u(y)\}.$$

Therefore u is quasiconcave.

The key step was not algebra. It was identifying the intermediate inequality that connected the assumption to the goal.

16 EXERCISE: FIND THE MISSING BRIDGE

For each problem, do not immediately write a complete proof.

Instead fill in:

$$\text{Assumption} \implies \boxed{\text{Intermediate statement}} \implies \text{Goal}.$$

16.1 Monotonic Transformation

Suppose $u : X \rightarrow \mathbb{R}$ represents a preference relation and

$$v(x) = g(u(x)),$$

where $g : \mathbb{R} \rightarrow \mathbb{R}$ is strictly increasing.

Show that

$$u(x) \geq u(y) \iff v(x) \geq v(y).$$

What is the key intermediate fact?

16.2 Convex Combination of Maximizers

Suppose f is concave and x^*, y^* are maximizers.

Show that every convex combination of x^* and y^* is also a maximizer.

What lower bound can concavity provide?

What upper bound follows from maximality?

16.3 Quadratic Form

Suppose A is positive definite.

Show that

$$x^\top Ax = 0 \implies x = 0.$$

What contradiction appears if $x \neq 0$?

17 A USEFUL EXPLORATION ROUTINE

When you understand a problem but do not yet see the proof, try the following questions.

1. Work forward.

What follows immediately from the assumptions?

2. Work backward.

What would be sufficient to establish the conclusion?

3. Look for a bridge.

Is there an intermediate statement that would connect the two?

4. Try examples.

What happens in a simple one-dimensional or numerical case?

5. Try to break the claim.

Can I find a counterexample?

6. Rewrite the statement.

Can I express the conclusion in an equivalent but more useful form?

7. Check unused assumptions.

Is there an assumption I have not used yet?

18 FINAL WORKSHOP I: UTILITY MAXIMIZATION

Consider a consumer problem

$$\max_{x \in B} u(x),$$

where $B \subseteq \mathbb{R}^n$ is convex and $u : B \rightarrow \mathbb{R}$ is strictly concave.

Suppose x^* and y^* both solve the problem.

Explore the claim:

$$x^* = y^*.$$

Do not write the final proof immediately.

1. Work forward from “ x^* and y^* are maximizers.”

What does this tell you about

$$u(x^*) \quad \text{and} \quad u(y^*)?$$

2. What does convexity of B allow you to construct?

3. Suppose $x^* \neq y^*$. What does strict concavity imply?

4. What contradiction would establish the result?

5. Write a proof skeleton.

x^*, y^* maximizers $\implies$ _____

B convex $\implies$ _____

u strictly concave $\implies$ _____

$\implies$ contradiction.

19 FINAL WORKSHOP II: IS THE CLAIM TRUE?

For each statement below, decide whether you should try to prove it or disprove it.

Do not assume that the claim is true.

19.1 Claim 1

If f and g are concave, then

$$fg$$

is concave.

19.2 Claim 2

If f and g are convex and $\alpha, \beta \geq 0$, then

$$\alpha f + \beta g$$

is convex.

19.3 Claim 3

If f is strictly concave, then every local maximizer is the unique global maximizer.

19.4 Claim 4

If A is positive definite, then all diagonal entries of A are positive.

20 TAKEAWAY

After deconstructing a problem, the next stage is exploration.

A useful mental model is



Your objective is to make the two sides meet.

When stuck, ask:

1. What follows from the assumptions?
2. What would be sufficient for the goal?
3. Is there an intermediate statement connecting them?
4. What happens in a simple example?
5. Can I construct a counterexample?
6. Can I rewrite the statement in a more useful form?
7. Which assumption have I not used yet?

A proof often looks obvious after it has been found.
The skill we are practicing is how to find it.